

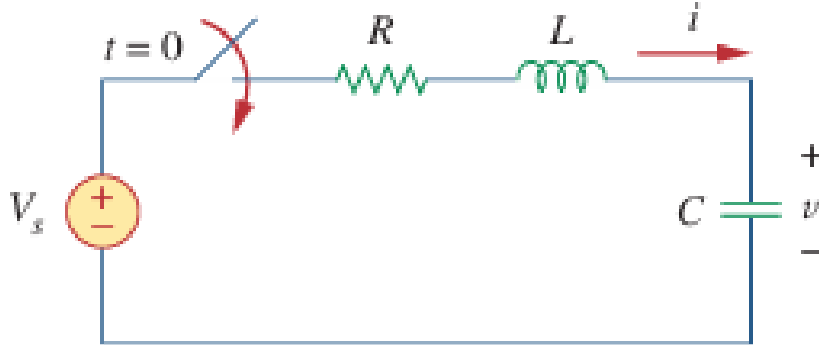
Module 2

Transient Analysis of First and Second-Order Circuits

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Step response of series RLC circuit



The step response is obtained by the sudden application of a dc source.

Consider no initial current through the inductor and initial voltage across the capacitor. The current through the inductor and initial voltage across the capacitor just after the switch is closed is the same as the initial condition.

Applying KVL around the loop for $t < 0$,
$$L \frac{di}{dt} + Ri + v = V_s \quad \text{--- 1}$$

But,
$$i = C \frac{dv}{dt} \quad \text{--- 2}$$

Substituting for i and rearranging,
$$\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC} \quad \text{--- 3}$$

The solution to Eqn. 3 has two components: the transient response $v_t(t)$ and the steady-state response $v_{ss}(t)$ that is,

$$v(t) = v_t(t) + v_{ss}(t)$$

Step response of series RLC circuit

$$v_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$v_t(t) = (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

--- 4

$$v_t(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

The steady-state response is the final value of $v(t)$. The final value of the capacitor voltage is the same as the source voltage V_s .

$$v_{ss}(t) = v(\infty) = V_s \quad \text{--- 5}$$

The complete solutions for the overdamped, underdamped, and critically damped cases are:

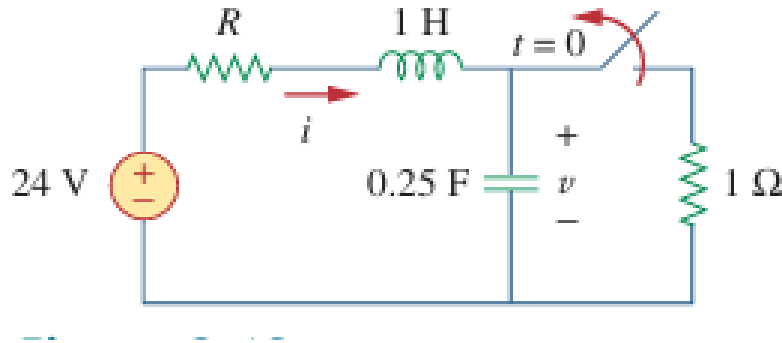
$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

Exercise

For the circuit, find $v(t)$ and $i(t)$ for $t > 0$. Consider the cases when $R = 5\ \Omega$, $R = 4\ \Omega$, and $R = 1\ \Omega$.



when $R = 5\ \Omega$, $t < 0$

the switch is closed for a long time. The capacitor behaves like an open circuit while the inductor acts like a short circuit. The initial current through the inductor is:

$$i(0) = \frac{24}{5 + 1} = 4\text{ A} \quad \text{--- 1}$$

The initial voltage across the capacitor is the same as the voltage across the $1\ \Omega$ resistor; that is,

$$v(0) = 1i(0) = 4\text{ V} \quad \text{--- 2}$$

Exercise

For $t > 0$, the switch is opened, so that we have the $1\ \Omega$ resistor disconnected. What remains is the series RLC circuit with the voltage source. The characteristic roots are determined as follows:

$$\alpha = \frac{R}{2L} = \frac{5}{2 \times 1} = 2.5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.25}} = 2$$
$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -1, -4 \quad \text{--- 3}$$

We find that $\alpha > \omega_0$ – we have an overdamped natural response. The total response is :

$$v(t) = v_{ss} + (A_1 e^{-t} + A_2 e^{-4t})$$

Where v_{ss} is the steady-state response. It is the final value of the capacitor voltage and $v_f = 24\text{ V}$

$$v(t) = 24 + (A_1 e^{-t} + A_2 e^{-4t}) \quad \text{--- 4}$$

We now need to find and using the initial conditions.

$$v(0) = 4 = 24 + A_1 + A_2$$

$$-20 = A_1 + A_2 \quad \text{--- 5}$$

Exercise

The current through the inductor cannot change abruptly and is the same current through the capacitor at $t = 0^+$ because the inductor and capacitor are now in series. Hence,

$$i(0) = C \frac{dv(0)}{dt} = 4 \quad \Rightarrow \quad \frac{dv(0)}{dt} = \frac{4}{C} = \frac{4}{0.25} = 16 \quad \text{--- 6}$$

Taking derivative of v in eqn.4,
$$\frac{dv}{dt} = -A_1 e^{-t} - 4A_2 e^{-4t} \quad \text{--- 7}$$

at $t = 0$,
$$\frac{dv(0)}{dt} = 16 = -A_1 - 4A_2 \quad \text{--- 8}$$

From eqn. 5 and eqn.8,
$$A_1 = -64/3 \text{ and } A_2 = 4/3.$$

eqn. 4 becomes,
$$v(t) = 24 + \frac{4}{3}(-16e^{-t} + e^{-4t}) \text{ V} \quad \text{--- 9}$$

Since the inductor and capacitor are in series for $t > 0$, the inductor current is the same as the capacitor current.
$$i(t) = C \frac{dv}{dt}$$

Multiplying eqn. 7 by $C = 0.25$ and substituting the values of A_1 and A_2 ,
$$i(t) = \frac{4}{3}(4e^{-t} - e^{-4t}) \text{ A}$$

Next class: Step response of parallel RLC circuit